

Generation Interconnection Request System Impact Study Report

For Queue Project Q11-0080

North Huntington 138 kV
November 24, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0080. The Interconnection Customer proposes a 106.0 MW solar photovoltaic facility interconnecting at the North Huntington 138 kV substation in the Virginia area. The request was received on November 02, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 70 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$20,529,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Parkin-Candlers Mtn 138.0 kV
Contingency: Wythe-South Creek 138.0 kV
Loading: 143.0% (Rating: 177 MVA)
DFAX: 5.2%
Pre-project loading: 139.8%
2. Facility: Monel-Candelaria 138.0 kV
Contingency: Clover-Bear 345.0 kV
Loading: 140.0% (Rating: 241 MVA)
DFAX: 6.2%
Pre-project loading: 137.3%
3. Facility: Desert Outpost-Monel 138.0 kV
Contingency: Brush Tavern-Rustburg 138.0 kV
Loading: 137.4% (Rating: 232 MVA)
DFAX: 5.6%
Pre-project loading: 134.8%
4. Facility: Desert Outpost-S Texa 138.0 kV
Contingency: Rustburg-Brush Tavern 138.0 kV
Loading: 133.4% (Rating: 319 MVA)
DFAX: 9.0%
Pre-project loading: 130.4%
5. Facility: Wythe-S Crk 138.0 kV
Contingency: Parkin-Candlers Mtn 138.0 kV Line
Loading: 127.4% (Rating: 195 MVA)
DFAX: 7.8%
Pre-project loading: 123.2%

6. Facility: Craven County Wood Energy Lp-Lone Tree 138.0 kV Line
Contingency: Bear-Clover 345.0 kV L
Loading: 121.6% (Rating: 226 MVA)
DFAX: 10.0%
Pre-project loading: 116.9%
7. Facility: S Lynchburg-Candlers Mtn 138.0 kV line
Contingency: S Crk Substation-Wythe 138.0 kV Line
Loading: 120.1% (Rating: 193 MVA)
DFAX: 5.2%
Pre-project loading: 117.3%
8. Facility: Craven County Wood Energy Lp-Lowmoor 138.0 kV
Contingency: Bear-Clover 345.0 kV
Loading: 117.5% (Rating: 230 MVA)
DFAX: 10.0%
Pre-project loading: 112.9%
9. Facility: Monel-Rustburg 138.0 kV
Contingency: Bear-Clover 345.0 kV
Loading: 115.8% (Rating: 292 MVA)
DFAX: 11.2%
Pre-project loading: 111.7%
10. Facility: S Texa-Franklin-Elk Grove 138.0 kV
Contingency: Desert Outpost-S Texa 138.0 kV Line
Loading: 114.8% (Rating: 239 MVA)
DFAX: 6.7%
Pre-project loading: 111.9%
11. Facility: Brush Tavern-Rustburg 138.0 kV
Contingency: Bear-Clover 345.0 kV
Loading: 112.5% (Rating: 331 MVA)
DFAX: 11.2%
Pre-project loading: 108.9%
12. Facility: Enslen-N Huntington 138.0 kV
Contingency: Florida-North Huntington 138.0 kV
Loading: 110.1% (Rating: 517 MVA)
DFAX: 52.1%
Pre-project loading: 99.5%
13. Facility: South Creek-Intel 138.0 kV Line
Contingency: Aibonito Pds-S Creek 138.0 kV
Loading: 109.6% (Rating: 191 MVA)
DFAX: 7.0%
Pre-project loading: 105.7%
14. Facility: Declo-New London 138.0 kV
Contingency: Fluvanna P.S Substation-Louisa Generation Facility 345.0 kV line
Loading: 109.6% (Rating: 257 MVA)
DFAX: 8.4%
Pre-project loading: 106.1%
15. Facility: Osceola-Lone Tree 138.0 kV
Contingency: Bear-Clover 345.0 kV
Loading: 109.5% (Rating: 263 MVA)
DFAX: 10.0%
Pre-project loading: 105.4%
16. Facility: Michigan Power Lp-Osceola 138.0 kV line
Contingency: Bear-Clover 345.0 kV

- Loading: 109.0% (Rating: 268 MVA)
DFAX: 10.0%
Pre-project loading: 105.0%
17. Facility: Chancellor Row-Bear Island 345.0 kV L
Contingency: Birchwood-Leavitt 345.0 kV
Loading: 108.5% (Rating: 1,105 MVA)
DFAX: 15.0%
Pre-project loading: 107.0%
18. Facility: Littlefork-Lake Lure 138.0 kV
Contingency: Louisa Generation Facility Substation-Fluvanna P.S 345.0 kV
Loading: 108.3% (Rating: 595 MVA)
DFAX: 6.8%
Pre-project loading: 107.1%
19. Facility: Sierra Pacific Lumber 138 kV/1 kV Transformer Bank
Contingency: Taber-Glenannan 345.0 kV
Loading: 107.7% (Rating: 528 MVA)
DFAX: 5.8%
Pre-project loading: 106.5%
20. Facility: Leavitt-Birchwood 345.0 kV line
Contingency: Bear Island-Dunn Siding 345.0 kV
Loading: 107.3% (Rating: 1,748 MVA)
DFAX: 17.2%
Pre-project loading: 106.3%
21. Facility: South Lynchburg-Brookville 138.0 kV
Contingency: S Crk-Wythe 138.0 kV
Loading: 107.2% (Rating: 207 MVA)
DFAX: 5.2%
Pre-project loading: 104.6%
22. Facility: South Colton-Yaphank 138.0 kV line
Contingency: Cristianitos-Enslin Substation 138.0 kV
Loading: 106.9% (Rating: 401 MVA)
DFAX: 9.4%
Pre-project loading: 104.5%
23. Facility: Boonsboro-Nacoochee 138.0 kV
Contingency: Brush Tavern-Rustburg 138.0 kV line
Loading: 106.8% (Rating: 354 MVA)
DFAX: 5.5%
Pre-project loading: 105.2%
24. Facility: Nacoochee-S Texa 138.0 kV
Contingency: Brush Tavern-Rustburg 138.0 kV
Loading: 106.5% (Rating: 358 MVA)
DFAX: 5.5%
Pre-project loading: 104.9%
25. Facility: Leavitt-Chancellor Row 345.0 kV Line
Contingency: Birchwood-Leavitt 345.0 kV
Loading: 105.7% (Rating: 1,220 MVA)
DFAX: 13.3%
Pre-project loading: 104.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Parkin Substation-Candlers Mtn 138.0 kV	Rebuild line with higher-capacity conductor	\$1,777,000	11% (\$197,000)
Monel-Candelaria 138.0 kV	Rebuild line with higher-capacity conductor	\$40,918,000	7% (\$2,928,000)
Monel-Desert Outpost 138.0 kV L	Rebuild line with higher-capacity conductor	\$4,301,000	10% (\$419,000)
Desert Outpost-S Texa 138.0 kV	Rebuild line with higher-capacity conductor	\$1,267,000	7% (\$88,000)
Wythe-S Crk 138.0 kV	Rebuild line with higher-capacity conductor	\$10,964,000	10% (\$1,080,000)
Lone Tree Substation-Craven County Wood Energy Lp 138.0 kV Line	Rebuild line with higher-capacity conductor	\$10,055,000	7% (\$701,000)
S Lynchburg-Candlers Mtn 138.0 kV Line	Rebuild line with higher-capacity conductor	\$2,477,000	11% (\$275,000)
Craven County Wood Energy Lp-Lowmoor 138.0 kV	Rebuild line with higher-capacity conductor	\$17,340,000	7% (\$1,210,000)
Monel-Rustburg 138.0 kV	Rebuild line with higher-capacity conductor	\$19,505,000	8% (\$1,572,000)
S Texa-Franklin-Elk Grove 138.0 kV	Rebuild line with higher-capacity conductor	\$18,132,000	6% (\$1,187,000)
Brush Tavern-Rustburg 138.0 kV	Rebuild line with higher-capacity conductor	\$9,794,000	8% (\$789,000)
Enslen-N Huntington 138.0 kV	Rebuild line with higher-capacity conductor	\$12,942,000	12% (\$1,498,000)
South Creek-Intel 138.0 kV Line	Reconductor line	\$4,294,000	10% (\$415,000)
Declo-New London 138.0 kV	Reconductor line	\$20,563,000	7% (\$1,462,000)
Osceola-Lone Tree 138.0 kV	Reconductor line	\$1,775,000	7% (\$124,000)
Michigan Power Lp-Osceola 138.0 kV	Reconductor line	\$5,172,000	7% (\$361,000)
Chancellor Row-Bear Island 345.0 kV L	Reconductor line	\$6,661,000	7% (\$445,000)
Littlefork-Lake Lure 138.0 kV	Reconductor line	\$2,000,000	7% (\$138,000)
Sierra Pacific Lumber 138-1 kV Transformer Bank	Replace transformer with higher-rated unit	\$4,204,000	7% (\$283,000)
Leavitt-Birchwood 345.0 kV	Reconductor line	\$65,261,000	7% (\$4,362,000)
S Lynchburg-Brookville 138.0 kV	Reconductor line	\$988,000	11% (\$110,000)
South Colton-Yaphank 138.0 kV line	Reconductor line	\$1,227,000	10% (\$122,000)
Boonsboro-Nacoochee 138.0 kV	Reconductor line	\$1,957,000	6% (\$108,000)
South Texa-Nacoochee 138.0 kV L	Reconductor line	\$1,348,000	6% (\$74,000)
Chancellor Row-Leavitt 345.0 kV	Reconductor line	\$8,694,000	7% (\$581,000)
	Total allocated to project		\$20,529,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.